

SEMINARARBEIT

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Ansible Workshop Protocol

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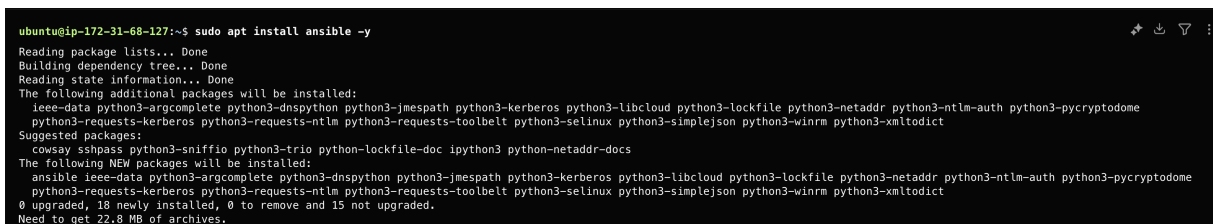
1 Ansible Workshop

In this workshop, we explore the capabilities of Ansible for automation and configuration management. We set up a master node and worker nodes, configure SSH access, and deploy playbooks to manage files and web servers.

1.1 Installation and Setup

1.1.1 Installing Ansible

First, we install Ansible on the master node. This involves updating the package manager and installing the ansible package.

A terminal window showing the command 'sudo apt install ansible -y' being executed on an Ubuntu system. The output shows the package lists being read, the dependency tree being built, and the state information being read. It then lists the additional packages to be installed, including 'ansible' and various Python libraries. The suggested packages are also listed. The final output shows the number of packages to be upgraded, newly installed, or not upgraded, and the total size of the archives to be downloaded.

```
ubuntu@ip-172-31-68-127:~$ sudo apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos python3-libcloud python3-lockfile python3-netaddr python3-ntlm-auth python3-pycryptodome
  python3-requests-kerberos python3-requests-ntlm python3-requests-toolbelt python3-selinux python3-simplejson python3-winrm python3-xlrd python3-xmldict
Suggested packages:
  cowsay sshpass python3-sniffio python3-trio python-lockfile-doc ipython3 python-netaddr-docs
The following NEW packages will be installed:
  ansible ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos python3-libcloud python3-lockfile python3-netaddr python3-ntlm-auth python3-pycryptodome
  python3-requests-kerberos python3-requests-ntlm python3-requests-toolbelt python3-selinux python3-simplejson python3-winrm python3-xlrd python3-xmldict
0 upgraded, 18 newly installed, 0 to remove and 15 not upgraded.
Need to get 22.8 MB of archives.
```

Abbildung 1: Installing Ansible on Master

1.1.2 SSH Key Generation

To allow passwordless authentication between the master and worker nodes, we generate an SSH key pair on the master node.

```
ubuntu@ip-172-31-68-127:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_rsa
Your public key has been saved in /home/ubuntu/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:dcrlKtvtetSHeutYAHYR4S/Kk7890+4QGUG+Qz1EBUU ubuntu@ip-172-31-68-127
The key's randomart image is:
+----[RSA 3072]-----+
|      .+...E      |
|      o  =      |
|      o  *  +      |
|      . + B  +      |
|      S  +      |
|      . + B.      |
|      *.+...      |
|      .o**      |
|      .o*B00      |
+----[SHA256]-----+

ubuntu@ip-172-31-68-127:~$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCdb5BCW3Jd01dTKcTpsH2Fpyp510q6GM0fCjAcELBtgH0nZq3MxGPiGlnIH2zk+LXR8cA260pAPS CWY6+FgS+4FAje+l+4FHzc9mLxEcQJYIvbaE+JFPoUe90/Z43I4xbwYH1oykgJY6FQRn8w
aV9xhatlPM6+5185hqlrKnnDqDzPyMHuVndYPWIKTSHMLZB3mn+EvWUt0f5CtcIM3x/Ek4t10p/c8hzMyB3jG914Y+e8o+3t/q267HRMimVM4nAZWd4arMAXcIxL1Cicr1uI1320ntcyx8m/erwXntFaWzRo6a2vRvFwcutFWK+UJ003vFv5is
b9uyyin/U80K4/qszpX9bk8q4hHtfeL4q0pFx2mjWDKfWu8gN0d2wtlBTesFh7Y5/nMacSpBHh09PrZ80evB8TeSj56c9CItP0V5mRq0f53lGZ0r/F3VVeatAPtJLrvwQQQYH1+eJIBe0lI0gttPFmYarDU08lBvnrKubvU0TKxPdpPoNe0tE=
ubuntu@ip-172-31-68-127
```

Abbildung 2: Generating SSH Keys

1.1.3 Distributing Keys

The public key is then copied to the worker nodes. This ensures that the master can connect to the workers without manual password entry.

```
ubuntu@ip-172-31-69-83:~$ cat >> ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCdb5BCW3Jd01dTKcTpsH2Fpyp510q6GM0fCjAcELBtgH0nZq3MxGPiGlnIH2zk+LXR8cA260pAPS CWY6+FgS+4FAje+l+4FHzc9mLxEcQJYIvbaE+JFPoUe90/Z43I4xbwYH1oykgJY6FQRn8w
aV9xhatlPM6+5185hqlrKnnDqDzPyMHuVndYPWIKTSHMLZB3mn+EvWUt0f5CtcIM3x/Ek4t10p/c8hzMyB3jG914Y+e8o+3t/q267HRMimVM4nAZWd4arMAXcIxL1Cicr1uI1320ntcyx8m/erwXntFaWzRo6a2vRvFwcutFWK+UJ003vFv5is
b9uyyin/U80K4/qszpX9bk8q4hHtfeL4q0pFx2mjWDKfWu8gN0d2wtlBTesFh7Y5/nMacSpBHh09PrZ80evB8TeSj56c9CItP0V5mRq0f53lGZ0r/F3VVeatAPtJLrvwQQQYH1+eJIBe0lI0gttPFmYarDU08lBvnrKubvU0TKxPdpPoNe0tE=
ubuntu@ip-172-31-68-127

ubuntu@ip-172-31-69-83:~$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGCdb5BCW3Jd01dTKcTpsH2Fpyp510q6GM0fCjAcELBtgH0nZq3MxGPiGlnIH2zk+LXR8cA260pAPS CWY6+FgS+4FAje+l+4FHzc9mLxEcQJYIvbaE+JFPoUe90/Z43I4xbwYH1oykgJY6FQRn8w
aV9xhatlPM6+5185hqlrKnnDqDzPyMHuVndYPWIKTSHMLZB3mn+EvWUt0f5CtcIM3x/Ek4t10p/c8hzMyB3jG914Y+e8o+3t/q267HRMimVM4nAZWd4arMAXcIxL1Cicr1uI1320ntcyx8m/erwXntFaWzRo6a2vRvFwcutFWK+UJ003vFv5is
b9uyyin/U80K4/qszpX9bk8q4hHtfeL4q0pFx2mjWDKfWu8gN0d2wtlBTesFh7Y5/nMacSpBHh09PrZ80evB8TeSj56c9CItP0V5mRq0f53lGZ0r/F3VVeatAPtJLrvwQQQYH1+eJIBe0lI0gttPFmYarDU08lBvnrKubvU0TKxPdpPoNe0tE=
ubuntu@ip-172-31-68-127
```

Abbildung 3: Saving Master Key on Worker

1.1.4 Verifying Connection

We verify the SSH connection to the worker nodes to ensure that the setup is correct.

```
ubuntu@ip-172-31-68-127:~$ ssh ubuntu@172.31.73.159
The authenticity of host '172.31.73.159 (172.31.73.159)' can't be established.
ED25519 key fingerprint is SHA256:J3uBYMwVD81yFoJNNUoi4Z8AocslgwE140XIfy7vmHg.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.73.159' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1043-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Dec  5 13:55:54 UTC 2025

System load:  0.08          Processes:      104
Usage of /:   23.5% of 7.57GB Users logged in:  1
Memory usage: 23%          IPv4 address for ens5: 172.31.73.159
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Dec  5 13:43:54 2025 from 195.245.225.253
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-73-159:~$ exit
logout
Connection to 172.31.73.159 closed.
```

Abbildung 4: Connection to Worker 1 Successful

```
ubuntu@ip-172-31-68-127:~$ ssh ubuntu@172.31.69.83
The authenticity of host '172.31.69.83 (172.31.69.83)' can't be established.
ED25519 key fingerprint is SHA256:UseIipZu9oP980pCvEG/vI+yz8AKKC4Lb0tkJvVY2ysE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.69.83' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1043-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Dec  5 13:56:33 UTC 2025

System load:  0.0          Processes:      104
Usage of /:   23.5% of 7.57GB Users logged in:  1
Memory usage: 26%          IPv4 address for ens5: 172.31.69.83
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Dec  5 13:45:21 2025 from 195.245.225.253
ubuntu@ip-172-31-69-83:~$ exit
logout
Connection to 172.31.69.83 closed.
```

Abbildung 5: Connection to Worker 2 Successful

1.2 Configuration

1.2.1 Creating Configuration

We create a directory for our Ansible project and generate a default configuration file.

```
ubuntu@ip-172-31-68-127:~$ mkdir ansible-lab

ubuntu@ip-172-31-68-127:~$ cd ansible-lab

ubuntu@ip-172-31-68-127:~/ansible-lab$ nano ansible.cfg

ubuntu@ip-172-31-68-127:~/ansible-lab$
```

Abbildung 6: Creating Ansible Config

1.2.2 Ansible Config File

The `ansible.cfg` file is customized to define the inventory file location and other defaults.

```
Master
GNU nano 6.2 ansible.cfg *
[defaults]
inventory = hosts
host_key_checking = False
```

Abbildung 7: Ansible Configuration File

1.2.3 Hosts Inventory

We define our inventory in the `hosts` file, listing the IP addresses of our worker nodes and grouping them if necessary.

```
Master
GNU nano 6.2 hosts *
[servers1]
server1 ansible_host=172.31.73.159

[servers2]
server2 ansible_host=172.31.69.83

[servers:children]
servers1
servers2
```

Abbildung 8: Hosts Inventory Configuration

1.2.4 Ping Test

We use the `ansible` command with the `ping` module to test connectivity to all hosts in our inventory.

```
ubuntu@ip-172-31-68-127:~/ansible-lab$ ansible all -m ping -o
server2 | SUCCESS => {"ansible_facts": {"discovered_interpreter_python": "/usr/bin/python3"},"changed": false,"ping": "pong"}
server1 | SUCCESS => {"ansible_facts": {"discovered_interpreter_python": "/usr/bin/python3"},"changed": false,"ping": "pong"}
```

Abbildung 9: Pinging Workers

1.3 Playbooks

1.3.1 File Operations

We start with a simple playbook to copy a file to the worker nodes.

```
ubuntu@ip-172-31-68-127:~/ansible-lab$ echo "This is a test file" > testfile.txt

ubuntu@ip-172-31-68-127:~/ansible-lab$ ls -la
total 20
drwxrwxr-x 2 ubuntu ubuntu 4096 Dec  5 14:15 .
drwxr-x--- 7 ubuntu ubuntu 4096 Dec  5 14:10 ..
-rw-rw-r-- 1 ubuntu ubuntu  55 Dec  5 14:07 ansible.cfg
-rw-rw-r-- 1 ubuntu ubuntu 130 Dec  5 14:13 hosts
-rw-rw-r-- 1 ubuntu ubuntu  20 Dec  5 14:15 testfile.txt

ubuntu@ip-172-31-68-127:~/ansible-lab$ ansible servers1 -m copy -a "src=testfile.txt dest=/home/ubuntu/testfile.txt mode=600"
server1 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "checksum": "b56df8ed5365fca1419818aa384ba3b5e7756047",
  "dest": "/home/ubuntu/testfile.txt",
  "gid": 1000,
  "group": "ubuntu",
  "md5sum": "5dd39cab1c53c2c77cd352983f9641e1",
  "mode": "0600",
  "owner": "ubuntu",
  "size": 20,
  "src": "/home/ubuntu/.ansible/tmp/ansible-tmp-1764944139.550931-3815-172802266990579/source",
  "state": "file",
  "uid": 1000
}
```

Abbildung 10: Creating Copy Playbook

We run the playbook and observe the output to ensure the task is changed.

```

ubuntu@ip-172-31-68-127:~/ansible-lab$ ansible all -m copy -a "src=testfile.txt dest=/home/ubuntu/testfile.txt mode=600"
server2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "checksum": "b56df8ed5365fca1419818aa384ba3b5e7756047",
  "dest": "/home/ubuntu/testfile.txt",
  "gid": 1000,
  "group": "ubuntu",
  "mode": "0600",
  "owner": "ubuntu",
  "path": "/home/ubuntu/testfile.txt",
  "size": 20,
  "state": "file",
  "uid": 1000
}
server1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "checksum": "b56df8ed5365fca1419818aa384ba3b5e7756047",
  "dest": "/home/ubuntu/testfile.txt",
  "gid": 1000,
  "group": "ubuntu",
  "mode": "0600",
  "owner": "ubuntu",
  "path": "/home/ubuntu/testfile.txt",
  "size": 20,
  "state": "file",
  "uid": 1000
}

```

Abbildung 11: Running Copy Playbook

We verify on the worker node that the file has indeed been copied.

```

ubuntu@ip-172-31-73-159:~$ ls -la
total 44
drwxr-x--- 6 ubuntu ubuntu 4096 Dec  5 14:15 .
drwxr-xr-x 3 root   root   4096 Dec  5 13:13 ..
drwx----- 3 ubuntu ubuntu 4096 Dec  5 14:11 .ansible
-rw----- 1 ubuntu ubuntu  174 Dec  5 13:56 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Jan  6  2022 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Jan  6  2022 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Dec  5 13:43 .cache
drwxrwxr-x 3 ubuntu ubuntu 4096 Dec  5 13:53 .local
-rw-r--r-- 1 ubuntu ubuntu  807 Jan  6  2022 .profile
drwx----- 2 ubuntu ubuntu 4096 Dec  5 13:54 .ssh
-rw----- 1 ubuntu ubuntu   20 Dec  5 14:15 testfile.txt

```

Abbildung 12: File Copied to Worker

We also create a playbook to delete the file and clean up.

```

ubuntu@ip-172-31-68-127:~/ansible-lab$ ansible all -m file -a "path=/home/ubuntu/testfile.txt state=absent"
server2 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "path": "/home/ubuntu/testfile.txt",
  "state": "absent"
}
server1 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": true,
  "path": "/home/ubuntu/testfile.txt",
  "state": "absent"
}

```

Abbildung 13: Deleting File Playbook

1.3.2 Webserver Deployment

Next, we create a more complex playbook to install and configure a web server (Apache/Nginx) on the workers.

```

Master
GNU nano 6.2 webserver.yaml *
---
- hosts: all
  become: true

  tasks:
    - name: Install latest version of Apache
      apt:
        name: apache2
        state: latest
        update_cache: yes

    - name: Replace standard index file in /var/www/html/index.html with custom index.html
      copy:
        src: index.html
        dest: /var/www/html/index.html

    - name: Restart Apache
      service:
        name: apache2
        state: restarted
      notify: Check HTTP Service

  handlers:
    - name: Check HTTP Service
      uri:
        url: http://{{ ansible_default_ipv4.address }}
        status_code: 200

```

Abbildung 14: Webserver Configuration Playbook

We first deploy this playbook to a single worker to test.

```
ubuntu@ip-172-31-68-127:~/ansible-labs$ echo "Hello World from Ansible" > index.html

ubuntu@ip-172-31-68-127:~/ansible-labs$ nano webserver.yaml

ubuntu@ip-172-31-68-127:~/ansible-labs$ ansible-playbook webserver.yaml --limit servers1

PLAY [all] *****
TASK [Gathering Facts] *****
ok: [server1]

TASK [Install latest version of Apache] *****
changed: [server1]

TASK [Replace standard index file in /var/www/html/index.html with custom index.html] *****
changed: [server1]

TASK [Restart Apache] *****
changed: [server1]

RUNNING HANDLER [Check HTTP Service] *****
ok: [server1]

PLAY RECAP *****
server1 : ok=5  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

Abbildung 15: Deploying to Worker 1

Once satisfied, we deploy it to all worker nodes.

```
ubuntu@ip-172-31-68-127:~/ansible-labs$ ansible-playbook webserver.yaml

PLAY [all] *****
TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]

TASK [Install latest version of Apache] *****
ok: [server1]
changed: [server2]

TASK [Replace standard index file in /var/www/html/index.html with custom index.html] *****
ok: [server1]
changed: [server2]

TASK [Restart Apache] *****
changed: [server1]
changed: [server2]

RUNNING HANDLER [Check HTTP Service] *****
ok: [server1]
ok: [server2]

PLAY RECAP *****
server1 : ok=5  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
server2 : ok=5  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
```

Abbildung 16: Deploying to All Workers

1.3.3 Verification

Finally, we verify the deployment by accessing the web server from a browser.

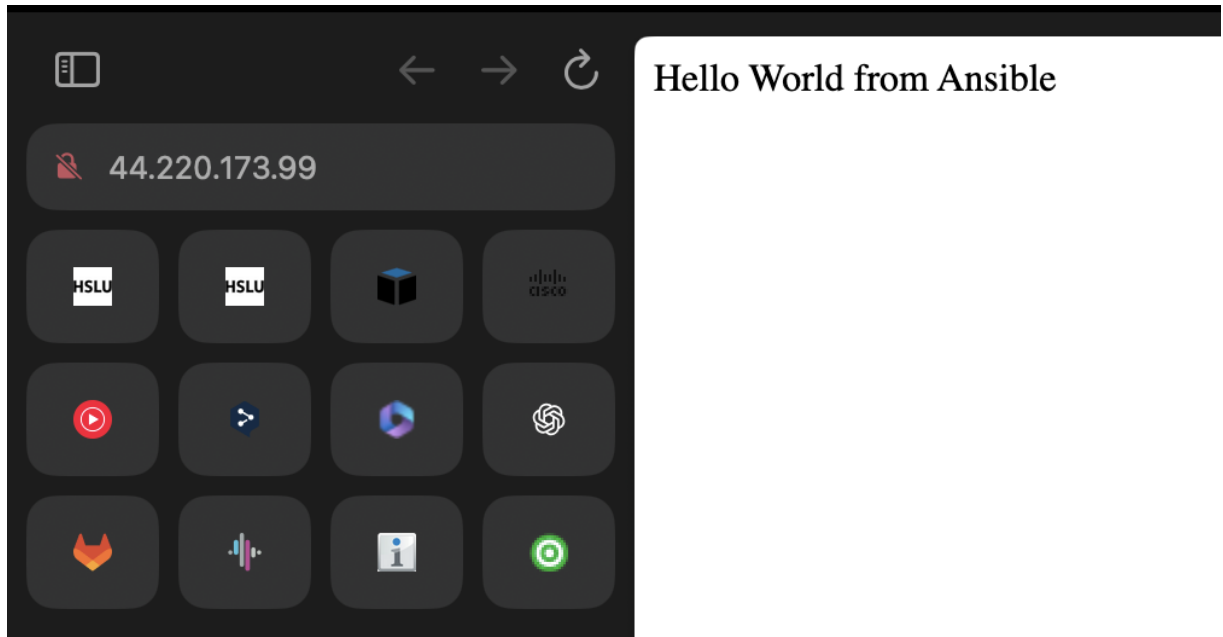


Abbildung 17: Website Successfully Deployed